

Introduction to Bayesian Data Analysis

Lecture 6: Nonlinear Models: GLMs and the Shifted Log-normal

Anne Giacobello & Julia Haaf

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Background

- GLMs are a big deal in frequentist stats
 - Do you use `glm` or `lme`?!?
- In Bayesian stats, we are used of thinking about our entire model more deliberately, including which model of the data to use.
- Yet, choosing between probability distributions for the data and interpreting the results requires a bit more thought than with linear models.

Overview

1. An Example

2. Model

- Generalized Linear Model
- Model of the Data
- Prior

3. Estimate the Model in `brms`

- Interpretation of the Results

4. Model fit

- Posterior prediction and Model comparison

5. The Shifted Log-normal

- Response times and the Shifted Log-normal Model
- Model in `brms`

An Example



Can mental health be predicted by daily habits and stress?

```
mentalhealth.dat <- read.csv("mental_health_dataset.csv")  
  
knitr::kable(head(mentalhealth.dat[, c(2,6,9,10,11,12)]))%>%  
  kable_styling(font_size = 16)
```

Age	Mental_Health_Condition	Stress_Level	Sleep_Hours	Work_Hours	Physical_Activity_Hours
36	No	Medium	7.1	46	5
34	Yes	Low	7.5	47	8
65	Yes	Low	8.4	58	10
34	No	Medium	9.8	30	2
22	Yes	Medium	4.9	62	5
64	Yes	High	6.3	34	0

An Example

"Do you have a mental illness?"



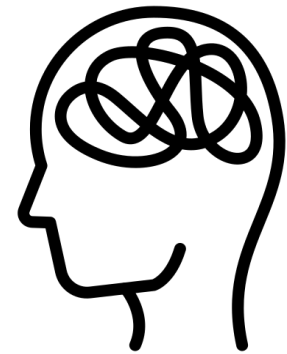
```
table(mentalhealth.dat$Mental_Health_Condition)
```

```
##
```

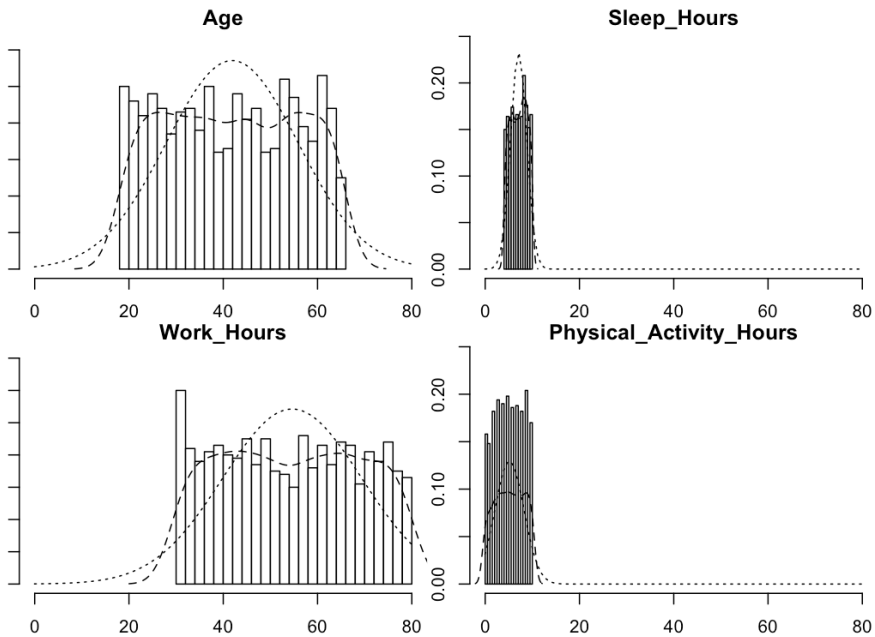
```
##   No  Yes
```

```
## 485 515
```

An Example



```
library("psych")  
multi.hist(mentalhealth.dat[, c(2, 10:12)])
```



The Model

What does a meaningful model for this data look like?

The Model

- Linear regression:

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2$$

- Extension:

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n$$

Problems?

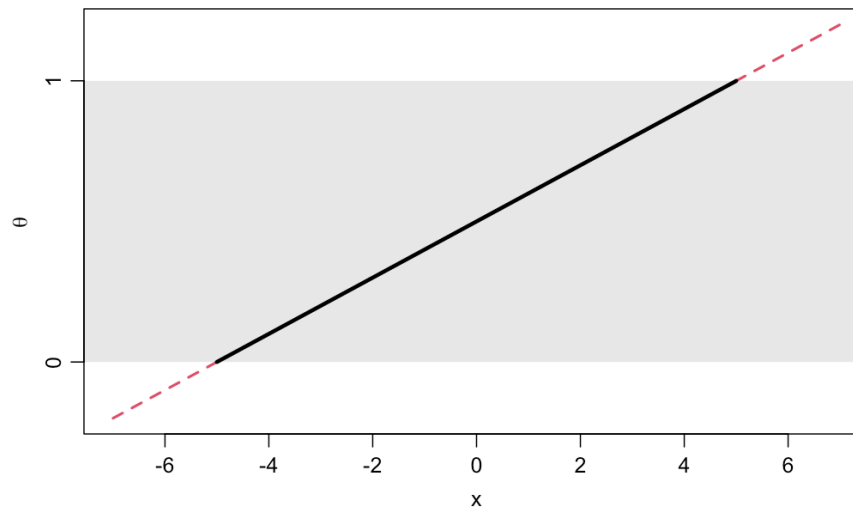
Y_i : Does the i th person have mental problems? $\rightarrow$ Yes/No
(1/0)

Statistical Model

Y_i : Does the i th person have mental problems? $\rightarrow$ Yes/No
(1/0)

- $Y_i \sim \text{Binomial}(n, \theta_i),$
- $0 > \theta > 1, n = ?$

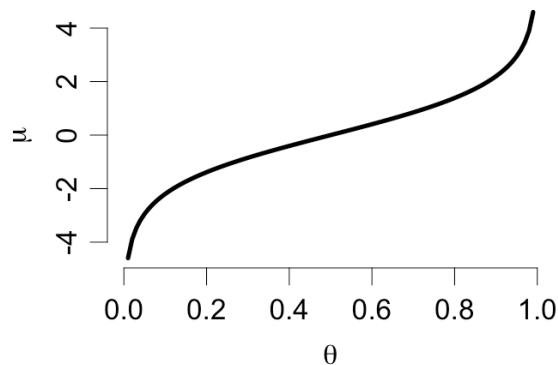
$$??? \theta_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n ???$$



Generalized Linear Model

Link Function $g(\cdot)$

- Goal: Connect the linear model with the parameter to be estimated (here: probability)
- For 0, 1 responses, the link function used is the logit transformation: $\mu_i = g(\theta_i) = \log \left(\frac{\theta_i}{1-\theta_i} \right)$



Generalized Linear Model

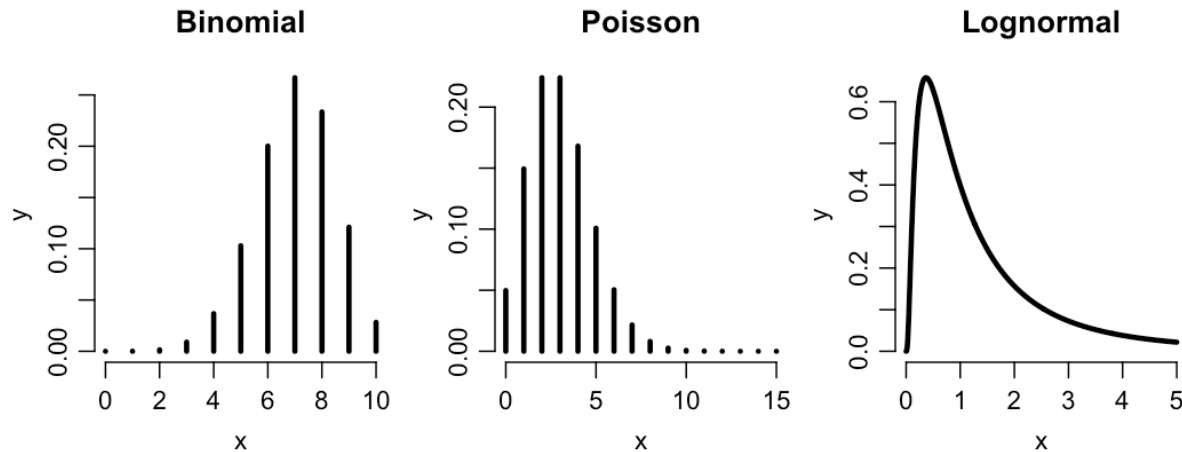
Using the link function, we go from the probability space into the "logit space" $\rightarrow \mu$ can be between $-\infty$ and ∞ .

Linear Model

$$\mu_i = \log \left(\frac{\theta_i}{1 - \theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n$$

Generalized Linear Model

The generalized linear model specifies the probability distribution of a random variable and a link function that allows flexible use of linear regression models.

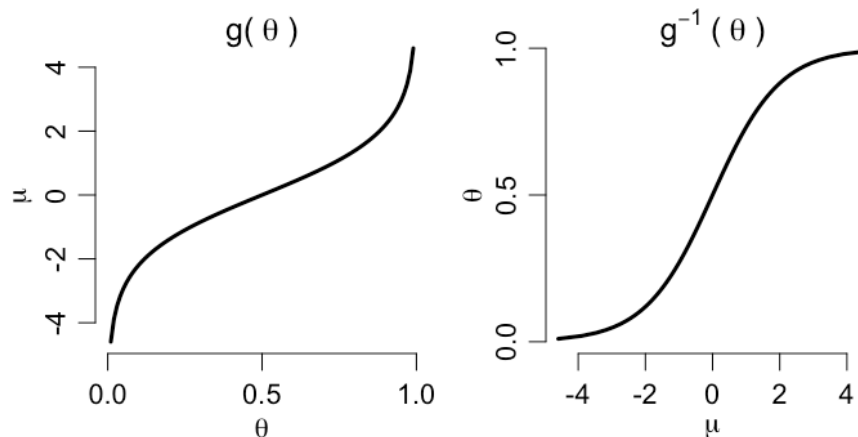


Generalized Linear Model

Inverse Function $g^{-1}(\cdot)$

- To get from μ_i back to θ_i we use the inverse of the logit function:

$$\theta_i = g^{-1}(\mu_i) = \frac{e^{\mu_i}}{1 + e^{\mu_i}}$$



Statistical Model

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- Predictors
 - $x_{i,1}$: Age
 - $x_{i,2}$: Stress level (high = 1, not high = 0)
 - $x_{i,3}$: Stress level (medium = 1, not medium = 0)
 - $x_{i,4}$: Sleep in hours
 - $x_{i,5}$: Work time per week in hours
 - $x_{i,6}$: Physical activity per week in hours
- $$\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$$

Statistical Model

How Does the Model Work?

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$
- Example:

$$\mu_i = -5 + x_{i,1}0.1 + x_{i,2}1 + x_{i,3}0.5 + x_{i,4}(-0.3) + x_{i,5}0.1 + x_{i,6}(-0.2)$$

Statistical Model

$$\mu_i = -5 + x_{i,1}0.1 + x_{i,2}1 + x_{i,3}0.5 + x_{i,4}(-0.3) + \\ x_{i,5}0.1 + x_{i,6}(-0.2)$$

- Age = 40
- Stress level = medium
- Sleep in hours = 7
- Work hours per week = 32
- Physical activity per week = 8

$$\mu_i = -5 + 40 \times 0.1 + 0 \times 1 + 1 \times 0.5 + 7 \times (-0.3) + \\ 32 \times 0.1 + 8 \times (-0.2) = -1$$

Statistical Model

$$\begin{aligned}\mu_i = & -5 + 40 \times 0.1 + 0 \times 1 + 1 \times 0.5 + 7 \times (-0.3) + \\ & 32 \times 0.1 + 8 \times (-0.2) = -1\end{aligned}$$

The probability of mental problems is then

$$\theta_i = g^{-1}(-1) = \frac{e^{-1}}{1+e^{-1}} = 0.27$$

Statistical Model

- Age = 50
- Stress level = high
- Sleep in hours = 5
- Work hours per week = 60
- Physical activity per week = 0

$$\mu_i = -5 + 50 \times 0.1 + 1 \times 1 + 0 \times 0.5 + 5 \times (-0.3) + \\ 60 \times 0.1 + 0 \times (-0.2) = 4.5$$

The probability of mental problems is then

$$\theta_i = g^{-1}(4.5) = \frac{e^{4.5}}{1+e^{4.5}} = 0.99$$

Prior

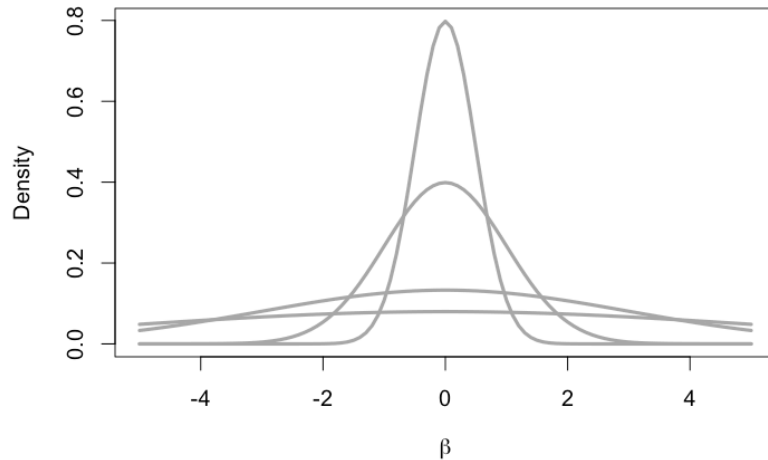
- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$

What parameters do we need in the model?

- All parameters!
 - $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6$
 - Normal distribution
- What about σ ?

Prior

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$



- $x_{i,1}$: Age
- $x_{i,2}$: Stress level high
- $x_{i,3}$: Stress level med
- $x_{i,4}$: Sleep in hours
- $x_{i,5}$: Work hours
- $x_{i,6}$: Physical activity

Prior

- $Y_i \sim \text{Binomial}(1, \theta_i),$
- $\mu_i = \log \left(\frac{\theta_i}{1-\theta_i} \right) = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,6}\beta_6$

```
library(brms)
```

```
# default_prior(Mental_Health_Condition ~ Age +  
#               Stress_Level + Sleep_Hours + Work_Hours  
#               + Physical_Activity_Hours  
#               , family = bernoulli(link = "logit")  
#               , data = mentalhealth.dat)
```

```
bprior <- c(prior(normal(0, 3), class = Intercept)  
            , prior(normal(0, 0.5), class = b))
```

Estimating the Model

Estimating the Model

```
model.1 <- brm(Mental_Health_Condition ~ Age +
               Stress_Level + Sleep_Hours + Work_Hours
               + Physical_Activity_Hours
               , family = bernoulli(link = "logit")
               , data = mentalhealth.dat
               , prior = bprior)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
##
```

```
## SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
```

```
## Chain 1:
```

```
## Chain 1: Gradient evaluation took 4.2e-05 seconds
```

```
## Chain 1: 1000 transitions using 10 leapfrog steps per transition
would take 0.42 seconds.
```

```
## Chain 1: Adjust your expectations accordingly!
```

```
## Chain 1:
```

```
## Chain 1:
```

```
## Chain 1: Iteration:      1 / 2000 [  0%]   (Warmup)
```

```
## Chain 1: Iteration:    200 / 2000 [ 10%]   (Warmup)
```

```
## Chain 1: Iteration:   1000 / 2000 [ 50%]   (Warmup)
```

Estimating the Model

```
summary(model.1)
```

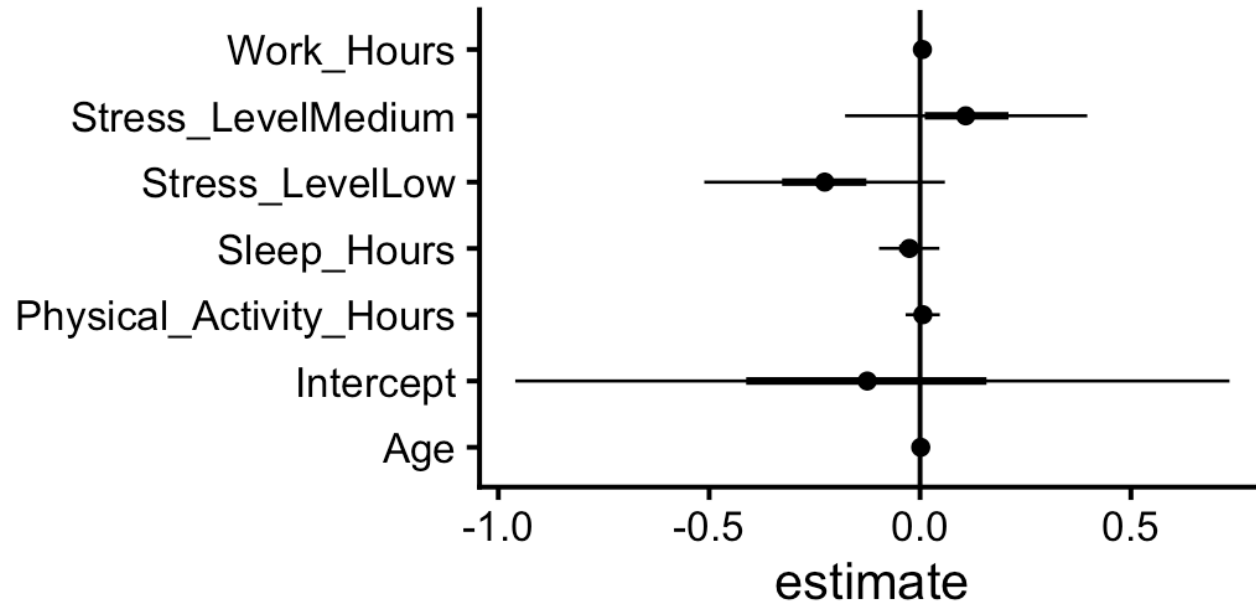
```
## Family: bernoulli
## Links: mu = logit
## Formula: Mental_Health_Condition ~ Age + Stress_Level + Sleep_Hours +
Work_Hours + Physical_Activity_Hours
## Data: mentalhealth.dat (Number of observations: 1000)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Regression Coefficients:
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Bulk_ESS					
## Intercept	-0.12	0.43	-0.96	0.73	1.00
6426					
## Age	0.00	0.00	-0.01	0.01	1.00
6359					
## Stress_LevelLow	-0.23	0.15	-0.51	0.06	1.00
4704					
## Stress_LevelMedium	0.11	0.15	-0.18	0.40	1.00
4591					
## Sleep_Hours	-0.03	0.04	-0.10	0.05	1.00
5396					

Estimating the Model

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.12	0.43	-0.96	0.73	1	6426.12	3577.71
Age	0.00	0.00	-0.01	0.01	1	6359.26	3323.19
Stress_LevelLow	-0.23	0.15	-0.51	0.06	1	4704.02	3142.33
Stress_LevelMedium	0.11	0.15	-0.18	0.40	1	4591.27	2967.32
Sleep_Hours	-0.03	0.04	-0.10	0.05	1	5395.68	3287.85
Work_Hours	0.01	0.00	0.00	0.01	1	7277.38	3328.61
Physical_Activity_Hours	0.01	0.02	-0.03	0.05	1	5396.86	3023.89

Estimating the Model

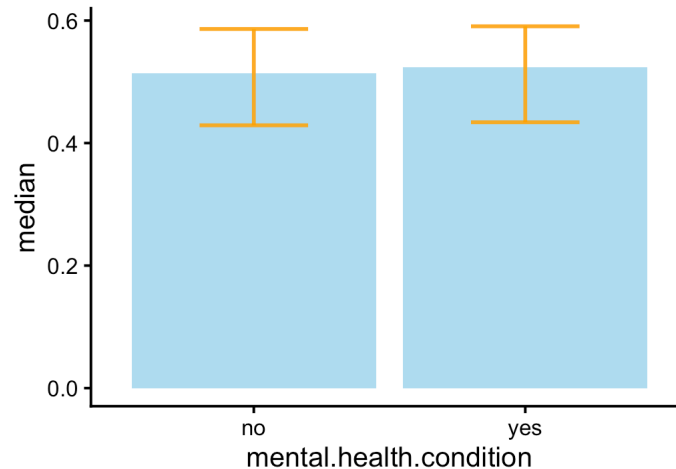


Posterior Prediction

Does the model accurately predict the response category for mental disorder?

```
post.pred <- posterior_predict(model.1) #as before
post.pred.mean <- colMeans(post.pred) #as before

# Median and prediction interval for y
post.pred.dat <- tapply(post.pred.mean
                        , mentalhealth.dat$Mental_Health_Condition
                        , quantile, probs = c(.025, .5, .975))
```



Posterior Prediction

```
# tapply gives a matrix, for ggplot we need a data frame
post.pred.dat <- as.data.frame(rbind(post.pred.dat$Yes
                                     , post.pred.dat$No))

# Naming the columns and rows
colnames(post.pred.dat) <- c("lower", "median", "upper")
post.pred.dat$mental.health.condition <- c("yes", "no")

ggplot(post.pred.dat) +
  geom_bar(aes(x = mental.health.condition
              , y = median)
          , stat = "identity"
          , fill = "skyblue"
          , alpha = 0.7) +
  geom_errorbar(aes(x = mental.health.condition
                   , ymin=lower
                   , ymax=upper)
               , width=0.4, colour="orange"
               , alpha=0.9, linewidth=1.3)+
  theme_classic(base_size = 20)
```

Model Comparison

Is the model informative overall?

→ Comparison with a model without predictors

```
model.1 <- brm(Mental_Health_Condition ~ Age +  
               Stress_Level + Sleep_Hours + Work_Hours  
               + Physical_Activity_Hours  
               , family = bernoulli(link = "logit")  
               , data = mentalhealth.dat  
               , prior = bprior  
               , save_pars = save_pars(all = TRUE)  
               , iter = 11000  
               , warmup = 1000  
               , silent = 2  
               , refresh = 0)
```

```
bprior <- c(prior(normal(0, 3), class = Intercept))
```

```
model.0 <- brm(Mental_Health_Condition ~ 1  
               , family = bernoulli(link = "logit")  
               , data = mentalhealth.dat  
               , prior = bprior
```

Model Comparison

```
bayes_factor(model.0, model.1)
```

```
## Warning: effective sample size cannot be calculated, has been  
replaced by
```

```
## number of samples.
```

```
## Iteration: 1
```

```
## Iteration: 2
```

```
## Iteration: 3
```

```
## Iteration: 4
```

```
## Iteration: 1
```

```
## Iteration: 2
```

```
## Iteration: 3
```

```
## Iteration: 4
```

```
## Estimated Bayes factor in favor of model.0 over model.1:  
1456674.79974
```

Model Comparison

- Strong evidence against effects of all predictors.
- Prevalence for a mental disorder cannot be predicted by age, life habits, and stress (for this dataset).
- Possible reasons?



Model Comparison

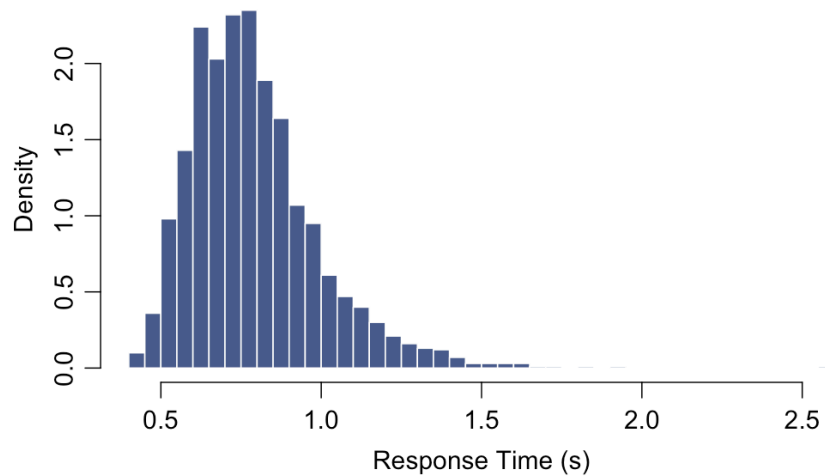
Take a closer look at the sample (slides 4-6): Work hours are from 30 to 80 hours/week, mental health split is close to 50/50, does this seem like the general population?

A GLM can only find effects that vary *within* the sample that it's been given. If the sample is already selected on factors related to the outcome, "no effect" maybe tells us more about the sampling process than about the relationship in the population.

The shifted log-normal model

Response times

Reaction times (RTs) are one of the most common outcome variables in psychology



Response times

What's so special?

- RTs are **always positive**
- The distribution is **right-skewed**: most responses cluster close together, but there is a long tail of slow responses that stretches to the right
- There appears to be a **minimum time** below which responses almost never occur

Response times

Why?

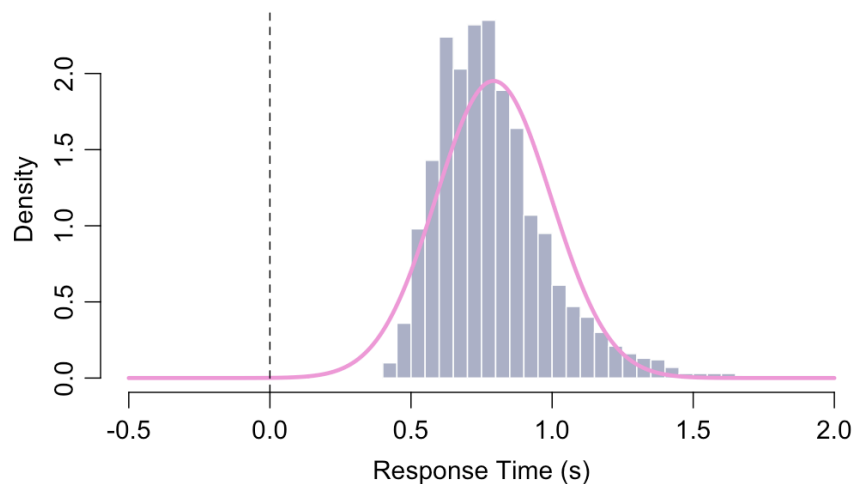
- A response can't happen before the stimulus has been perceived, processed, and a motor response prepared and executed, which means there is a **hard lower bound** (often called t_0 , encoding & motor time)
- Most trials are quick and similarly responded to but sometimes our attention lapses or motor execution is delayed (e.g., because we are tired), producing a **long right tail**

Response times

- Because of these features a **symmetric, unbounded** distribution (like the Normal) is a poor fit

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

would assign probability to $Y_i < 0$, and cannot reproduce the skew



The Shifted Log-normal

We need a distribution for the data that:

1. Puts mass only for values above this encoding time (τ or non-decision time)
2. And is right-skewed

$Y_i - \tau$ could be log-normally distributed. Then Y_i is bigger than τ . And since the log-normal is right-skewed, Y_i is right-skewed then as well.

This is the **shifted log-normal** model:

$$Y_i - \tau \sim \text{Log-Normal}(\mu_i, \sigma^2)$$

The Shifted Log-normal

Link Function

- Because we already are on the $\log(Y_i - \tau)$ scale, the link function is:

$$\mu_i = \beta_0 + x_{i,1}\beta_1 + x_{i,2}\beta_2 + \dots + x_{i,n}\beta_n$$

- μ_i is the (linear predictor for the) mean of the *log* RT above the shift and not the mean RT

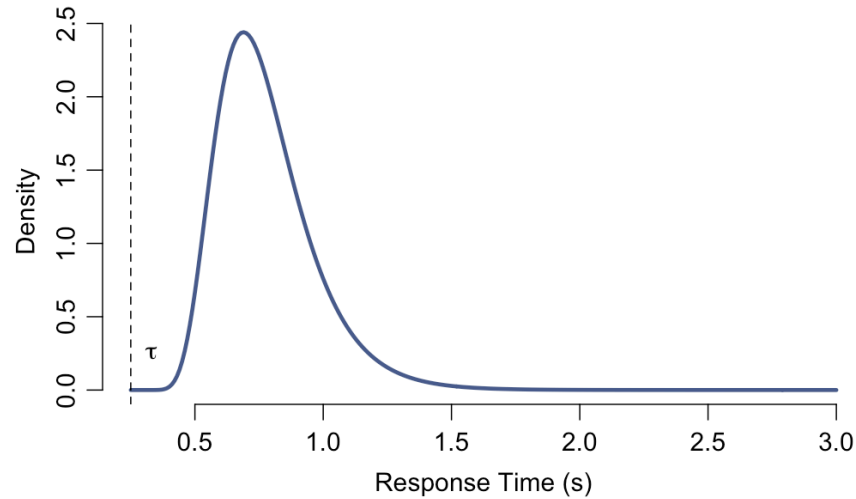
The Shifted Log-normal

- To get back to the RT scale, we need the to inverse transform:

$$\mathbb{E}[Y_i] = \tau + \exp\left(\mu_i + \frac{\sigma^2}{2}\right)$$

- τ is also a parameter that needs to be estimated, it has the constraint $\tau < \min(Y_i)$, since nobody can respond before the shift

The Shifted Log-normal



The shift τ moves the whole distribution so that it starts after encoding time.

The Shifted Log-normal

Statistical Model

- $Y_i - \tau \sim \text{Log-Normal}(\mu_i, \sigma^2)$
- $\mu_i = \beta_0 + x_{i,1}\beta_1 + \dots + x_{i,n}\beta_n$
- Parameters to estimate:
 - $\beta_0, \dots, \beta_n$ (linear predictor)
 - σ (residual SD on log scale)
 - τ (shift / non-decision time)

The Shifted Log-normal

- Priors still need to respect the parameter space:
 - Normal priors on the β s (it's a log-RT scale, so this should be small, e.g. $\beta \sim \text{Normal}(0, 0.5)$)
 - A positive prior on σ (e.g. half-Normal or Exponential)
 - A prior on τ bounded to below $\min(Y_i)$

The Shifted Log-normal

```
library(brms)

bprior <- c(prior(normal(0, 3), class = Intercept)
            , prior(normal(0, 0.5), class = b)
            , prior(normal(0, 1), class = sigma)
            , prior(uniform(0, min_RT), class = ndt, lb = 0, ub =
min_RT))

model.rt <- brm(RT ~ Age + Stress_Level + Sleep_Hours + Work_Hours
                + Physical_Activity_Hours
                , family = shifted_lognormal()
                , data = rt.dat
                , prior = bprior)
```

`brms` calls the shift parameter `ndt` (non-decision time), that's τ .

The Shifted Log-normal

Disclaimer

- We are tying the prior to `min_RT` and that means the prior bound depends on this particular sample
- In practice, RT data are instead usually cleaned before modeling
- So we would be excluding trials faster than some fixed threshold for example (often around 150–200 ms) since responses that are this fast cannot reflect real stimulus processing
- If we have already excluded everything below 200 ms, you *know* a priori that τ must be below that cutoff

The Shifted Log-normal

Interpretation

- Coefficients β_j are on the **log-RT** scale: one unit increase in x_j multiplies the (shifted) RT by $\exp(\beta_j)$
- $\exp(\beta_j) > 1$: predictor is associated with **slower** responses
- $\exp(\beta_j) < 1$: predictor is associated with **faster** responses
- τ (**ndt**) is interpretable in seconds *here*, it's the estimated bound below which no response is possible

The Shifted Log-normal

caution: `ndt` in `brms`

- `brms` defines the shifted log-normal with default links `mu = identity`, `sigma = log`, and `ndt = log`
- in our example, τ is in seconds because we made it a constant (no formula) and gave it lower and upper bounds, this tells `brms` to keep it on the RT scale --> you can check this with `make_stancode()`
- If you want to let `ndt` vary by predictors or by subject/item (e.g. `ndt ~ 1 + (1 | subj)`), `brms` applies the default `log` --> the reported estimates and the priors you put on them are on the `log` scale then and not in seconds

See this [Stan Forums thread](#) for an example, the modeled `ndt` intercept comes out as -23 .

